

TITAN SMALL BUSINESS SERVICES: SECURITY ARCHITECTURE DOCUMENT (SAD)

Operator: Reshad **Date:** 4/20/26

1. Perimeter Hardening (UFW & SSH)

- **SSH Status:** I edited the SSH config and turned off root login and password login. Then I restarted SSH.
- **Firewall Logic:** I set the firewall to block incoming traffic by default. I only allowed port 22 for SSH and port 8080 for the web app

2. The Automated Auditor (Python)

- **Script Logic:** My Python script uses the `os` module to ping the Windows DC 4 times. It writes `DC is UP` or `DC is DOWN` to the log file depending on the result.
- **Telemetry Path:** `/var/log/sys_audit.log`

3. Containerized App (Docker)

- **Network Isolation:** I made two Docker networks called `frontend` and `backend`. The wiki container is on port 8080, and the database stays on the backend network so it is not exposed to the host.
- **Stack Health:** Both containers started correctly. I tested `curl http://localhost:8080` and got the nginx page back.

4. Executive Summary

I hardened the Ubuntu system by locking down SSH and setting firewall rules. I also created a Python audit script and deployed the Docker app with the database isolated from the host. After testing everything, the system was working and the frontend was reachable on port 8080.